# DATA SCIENCE PROJECT REPORT

# STUDENT BEHAVIOR ANALYSIS


## Kumari Sneha
## Roll No: 40
## Section : 324PB
## Lovely Professional University


### Certificate

This is to certify that Kumari Sneha has completed the project titled 'Student Behavior Analysis Using Machine Learning' under my guidance.

### Declaration

I, **Kumari Sneha**, a student of **B.Tech Computer Science and Engineering** at **Lovely Professional University, Punjab**, hereby declare that the project titled **"College Student Analysis"** submitted in partial fulfillment of the requirements for the award of the degree is a record of original work carried out by me under the guidance and supervision of the concerned faculty.

I further declare that the work presented in this project report is based on my own efforts, research, and understanding. All the data analysis, visualizations, and interpretations have been performed by me using appropriate tools and techniques in the field of Data Science. Wherever references have been made to the work of others, due acknowledgment has been given in accordance with standard academic practices.

I confirm that this project report has not been submitted, either in part or in full, to any other university or institution for the award of any degree, diploma, or certification. The results and findings presented in this report are genuine and have not been fabricated or manipulated in any form.

I also take full responsibility for maintaining the integrity and authenticity of the work. I understand that any form of plagiarism or academic misconduct will lead to disciplinary action as per the rules and regulations of the university.

This project has enhanced my understanding of data analysis, visualization techniques, and machine learning concepts. It has provided me with practical exposure to real-world data and strengthened my problem-solving and analytical skills.

I hereby declare that the information provided in this report is true to the best of my knowledge and belief.

## Acknowledgement

I would like to express my sincere gratitude to all those who have supported and guided me throughout the completion of this project titled **"College Student Analysis."**

I am especially thankful to my project guide and faculty members for their valuable guidance, continuous encouragement, and constructive suggestions, which helped me successfully complete this work.

I would also like to thank **Lovely Professional University** for providing me with the necessary resources, infrastructure, and learning environment to carry out this project effectively.

I extend my heartfelt thanks to my friends and peers for their support, cooperation, and helpful discussions during the course of this project.

Lastly, I am deeply grateful to my family for their constant motivation, understanding, and encouragement, which enabled me to complete this project successfully.

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## Introduction

In today's data-driven world, the analysis of student-related data plays a crucial role in improving academic performance, institutional planning, and decision-making processes. Educational institutions generate a vast amount of data related to student demographics, academic records, attendance, and extracurricular activities. Analyzing this data can provide valuable insights into student behavior, performance patterns, and factors influencing academic success.

The project titled **"College Student Analysis"** focuses on exploring and analyzing student data using data science techniques. The primary objective is to identify meaningful patterns, trends, and relationships within the dataset that can help in understanding student performance and engagement. By applying data preprocessing, exploratory data analysis (EDA), and visualization techniques, the project aims to transform raw data into actionable insights.

This study utilizes tools such as Python and its libraries (like Pandas, NumPy, Matplotlib, and Seaborn) to clean, process, and visualize the data effectively. Through graphical representations such as bar charts, histograms, and correlation plots, the project highlights key findings that can assist educators and administrators in making informed decisions.

Furthermore, this analysis helps in identifying factors that may impact student outcomes, such as study habits, attendance, and other academic indicators. The insights derived from this project can be useful in improving teaching strategies, enhancing student support systems, and ultimately contributing to better academic results.

In conclusion, this project demonstrates the importance of data analysis in the education sector and showcases how data-driven approaches can lead to improved understanding and optimization of student performance.

## Scope of the Analysis

The scope of this project is centered around the analysis of college student data to extract meaningful insights related to academic performance and behavioral patterns. The analysis primarily focuses on understanding how different factors such as attendance, study time, and other academic indicators influence student outcomes.

This project involves data preprocessing, including handling missing values, data cleaning, and transformation to ensure the dataset is suitable for analysis. Exploratory Data Analysis (EDA) is performed to identify trends, patterns, and relationships among various attributes in the dataset. Visualization techniques are used to represent the findings in an intuitive and understandable manner.

The scope is limited to descriptive and exploratory analysis using statistical and visualization methods. It does not extend to advanced predictive modeling or deployment of machine learning models (unless explicitly included). The results obtained are based on the given dataset and may vary with different datasets or real-world scenarios.

Additionally, this project aims to provide insights that can assist educators and institutions in understanding student performance better. However, it does not make definitive conclusions about all student populations, as the analysis is constrained by the size, quality, and nature of the dataset used.

In summary, the project emphasizes data-driven understanding of student-related information within a defined dataset, focusing on analysis, interpretation, and visualization rather than real-time implementation or large-scale system development.

## Existing System

In the current educational environment, most institutions rely on traditional methods for managing and analyzing student data. These methods primarily include manual record-keeping, spreadsheets, and basic database systems. While these approaches allow institutions to store large amounts of data, they often lack the capability to extract meaningful insights efficiently.

Existing systems typically focus on data storage rather than data analysis. For example, student information systems (SIS) maintain records such as attendance, grades, and personal details, but they provide limited analytical features. As a result, educators and administrators must manually interpret the data, which can be time-consuming and prone to errors.

Moreover, traditional systems do not effectively utilize modern data science techniques such as exploratory data analysis, visualization, and pattern recognition. This limits the ability to identify trends, correlations, and hidden patterns that could help in improving student performance and decision-making.

Another limitation of the existing system is the lack of interactive and visual representations. Data is often presented in tabular form, making it difficult to quickly understand complex relationships or derive actionable insights. Additionally, these systems may not handle missing or inconsistent data efficiently, leading to inaccurate analysis.

In summary, while the existing systems are useful for data storage and basic management, they fall short in providing advanced analytical capabilities. This highlights the need for a more efficient, data-driven approach that can transform raw student data into meaningful and actionable insights.

## Drawbacks of Existing System

The existing systems used for managing and analyzing student data have several limitations that reduce their effectiveness in deriving meaningful insights. Some of the major drawbacks are as follows:

- **Lack of Advanced Analysis:**
  Traditional systems mainly focus on data storage and basic operations. They do not support advanced analytical techniques such as data mining, pattern recognition, or predictive analysis.

- **Manual Effort and Time-Consuming:**
  A significant amount of manual effort is required to interpret data, which makes the process slow and inefficient. This can delay decision-making.

- **Limited Data Visualization:**
  Data is mostly presented in tabular form, making it difficult to understand trends, patterns, and relationships quickly.

- **Higher Chances of Errors:**
  Manual handling and interpretation of data increase the risk of human errors, leading to inaccurate conclusions.

- **Poor Handling of Large Data:**
  Traditional methods may not efficiently handle large datasets, which can affect performance and scalability.

- **Lack of Real-Time Insights:**
  Existing systems do not provide real-time analysis, making it difficult to take timely actions based on current data.

- **Data Inconsistency Issues:**
  Missing, duplicate, or inconsistent data is often not handled properly, which affects the quality of analysis.
- **Limited Decision Support:**
  These systems do not provide actionable insights or recommendations, making it harder for educators to make informed decisions.

## Source of Dataset

The dataset used in this project has been sourced from **Kaggle**, an online platform that provides a wide range of publicly available datasets for data science, machine learning, and research purposes.

Kaggle datasets are widely recognized for their quality, diversity, and real-world relevance. The dataset used in this project contains information related to college students, including various academic and behavioral attributes that are suitable for analysis.

Before performing the analysis, the dataset was downloaded from Kaggle and preprocessed to ensure accuracy and consistency. Data cleaning steps such as handling missing values, removing duplicates, and formatting were carried out to prepare the dataset for further analysis.

The dataset is used strictly for **academic and educational purposes**, and no sensitive or personal information has been misused. Proper care has been taken to maintain the integrity and reliability of the data throughout the project.

## ETL Process

The ETL (Extract, Transform, Load) process is a fundamental step in data analysis that involves collecting data from sources, preparing it for analysis, and storing it in a structured format. In this project, the ETL process was implemented as follows:

### 4.1 Extract

The first step involved extracting the dataset from **Kaggle**, a reliable online data repository. The dataset was downloaded in a structured format (such as CSV) and imported into the Python environment using libraries like **Pandas**. This step ensured that the raw data was readily available for further processing.

### 4.2 Transform

The transformation phase focused on cleaning and preparing the data to make it suitable for analysis. The following operations were performed:

- Handling missing or null values using appropriate techniques (removal or imputation)
- Removing duplicate records to maintain data integrity
- Converting data types into appropriate formats
- Filtering irrelevant or inconsistent data
- Structuring the dataset for better readability and analysis

Additionally, exploratory data analysis (EDA) techniques were applied to understand the distribution and relationships within the data.

### 4.3 Load

After transformation, the cleaned and processed dataset was loaded into a suitable format for analysis and visualization. The data was used within the Python environment for generating insights, visualizations, and results. In some cases, the processed data can also be stored in files or databases for future use.

The ETL process ensured that the raw dataset was converted into a clean, structured, and analysis-ready format, enabling accurate and meaningful insights in the project.

## Analysis on Dataset

The analysis of the dataset is a crucial part of this project, as it helps in understanding the underlying patterns, relationships, and trends present in student data. The primary objective of this analysis is to explore how different academic, behavioral, and lifestyle factors influence student performance, particularly in terms of GPA.

The dataset used in this project contains multiple variables such as study habits, sleep patterns, stress levels, social media usage, phone usage, and other relevant attributes. Each of these factors plays a significant role in determining the academic outcomes of students. By applying data analysis techniques, meaningful insights can be derived that can help in understanding student behavior and improving performance.

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### 6.2 General Description of Dataset

The dataset consists of structured data where each row represents an individual student, and each column represents a specific feature or attribute. The features include both numerical and categorical variables, making the dataset suitable for comprehensive analysis.

Key attributes in the dataset include:

- Study-related variables (e.g., study sessions)
- Lifestyle factors (e.g., sleep hours, phone usage, social media usage)
- Psychological factors (e.g., stress levels)
- Academic performance indicator (GPA)

The dataset provides a holistic view of student life, allowing for the analysis of both academic and non-academic factors. This makes it possible to study not only direct influences but also indirect relationships between variables.

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### 6.3 Data Preprocessing

Before performing the analysis, the dataset was preprocessed to ensure accuracy and consistency. Data preprocessing is a critical step as raw data often contains missing values, inconsistencies, or irrelevant information.

The following preprocessing steps were carried out:

- **Handling Missing Values:** Missing or null values were identified and treated appropriately using techniques such as removal or imputation.
- **Removing Duplicates:** Duplicate records were eliminated to avoid redundancy and ensure data integrity.
- **Data Cleaning:** Irrelevant or inconsistent entries were filtered out to improve data quality.
- **Data Transformation:** Certain variables were transformed into suitable formats for analysis, such as categorizing continuous variables into meaningful groups (e.g., phone usage levels).

These steps ensured that the dataset was clean, reliable, and ready for further analysis.

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### 6.4 Specific Requirements, Functions and Formulas

Various data analysis techniques and functions were used to perform the analysis. Python libraries such as **Pandas, NumPy, Matplotlib, and Seaborn** were utilized extensively.

Some of the key operations performed include:

- **Descriptive Statistics:** Used to summarize data using measures such as mean, median, and standard deviation.
- **Correlation Analysis:** Used to determine relationships between variables using correlation coefficients.
- **Data Visualization:** Graphical representations such as heatmaps, scatter plots, bar charts, and box plots were used to interpret data.
- **Categorization Functions:** Techniques like binning were used to group continuous variables (e.g., phone usage levels).

These tools and techniques helped in simplifying complex data and extracting meaningful insights.

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### 6.5 Analysis Results

The analysis revealed several important patterns and relationships within the dataset:

- A strong positive relationship was observed between **study-related factors and GPA**, indicating that consistent academic effort leads to better performance.
- **Sleep patterns** showed a moderate positive impact on performance, highlighting the importance of maintaining a healthy routine.
- **Stress levels** were found to have a negative effect on GPA, suggesting that higher stress reduces academic efficiency.
- **Phone usage and social media usage** showed a negative relationship with GPA, indicating that excessive digital engagement can lead to distractions.
- **Balanced involvement in extracurricular activities** was found to be beneficial, while excessive involvement may negatively impact performance.

These results demonstrate that student performance is influenced by a combination of academic habits, lifestyle choices, and psychological factors.

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### 6.6 Visualization (Dashboard)

Visualization played a vital role in this project by converting raw data into meaningful graphical representations. Various plots such as **heatmaps, scatter plots, bar charts, box plots, and pair plots** were used to analyze and interpret the data.

The visualizations helped in:

- Identifying trends and patterns easily
- Understanding relationships between variables
- Detecting outliers and inconsistencies
- Presenting findings in a clear and intuitive manner

For example, heatmaps were used to show correlations, scatter plots to analyze relationships, and bar charts to compare different categories. These visual tools made it easier to communicate the results effectively.

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### 6.7 Summary

Overall, the dataset analysis provided valuable insights into the factors affecting student performance. It highlighted the importance of maintaining a balance between academic efforts and lifestyle habits. The use of data preprocessing, statistical analysis, and visualization techniques ensured that the results were accurate and meaningful.

This section forms the foundation for further analysis and machine learning implementation, ultimately contributing to a better understanding of student behavior and performance.

## Correlation Heatmap Analysis

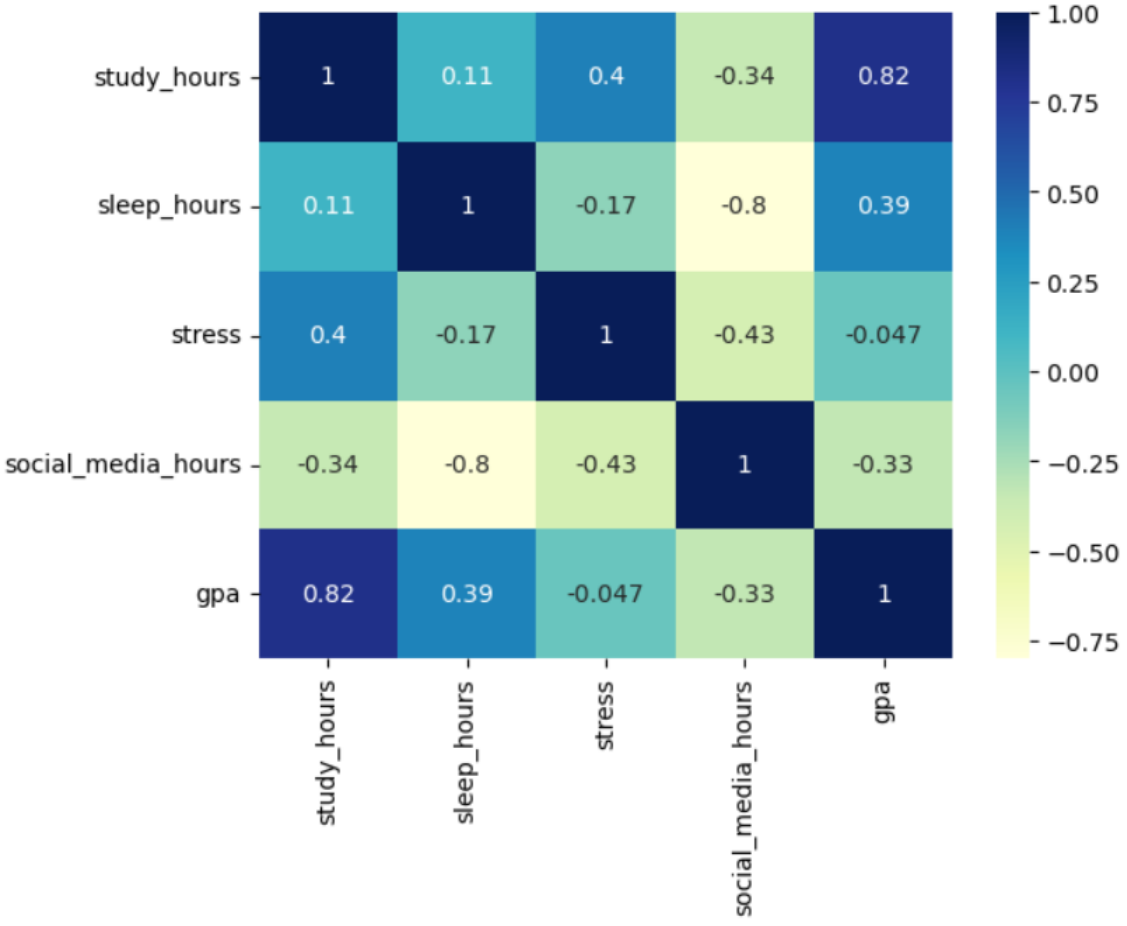
The correlation heatmap is one of the most important visualizations used in this project, as it provides a comprehensive overview of the relationships between different variables in the dataset. In this analysis, a heatmap was generated using variables such as study hours, sleep hours, stress levels, social media usage, and GPA. The purpose of this plot was to identify how strongly each factor is related to academic performance.

From the heatmap, it was observed that **study hours have a strong positive correlation with GPA**, indicating that students who dedicate more time to studying tend to achieve

better academic results. Similarly, **sleep hours also show a moderate positive relationship with GPA**, suggesting that adequate rest contributes to improved cognitive performance and learning ability.

On the other hand, **stress levels exhibit a negative correlation with GPA**, meaning that students experiencing higher stress tend to perform worse academically. This highlights the importance of mental well-being in academic success. Additionally, **social media usage shows a slight negative correlation with GPA**, indicating that excessive time spent on digital platforms may distract students from their studies.

Overall, the heatmap provides a clear understanding of how different lifestyle and behavioral factors interact with academic performance. It serves as a foundation for further analysis and confirms that both academic habits and personal well-being significantly influence student outcomes




## Phone vs GPA

The relationship between phone usage and academic performance was analyzed to understand the impact of digital device usage on students' GPA. In this analysis, the

amount of time students spent on their phones was considered as a key variable, and its effect on GPA was examined using visualization techniques such as bar charts and categorized comparisons.

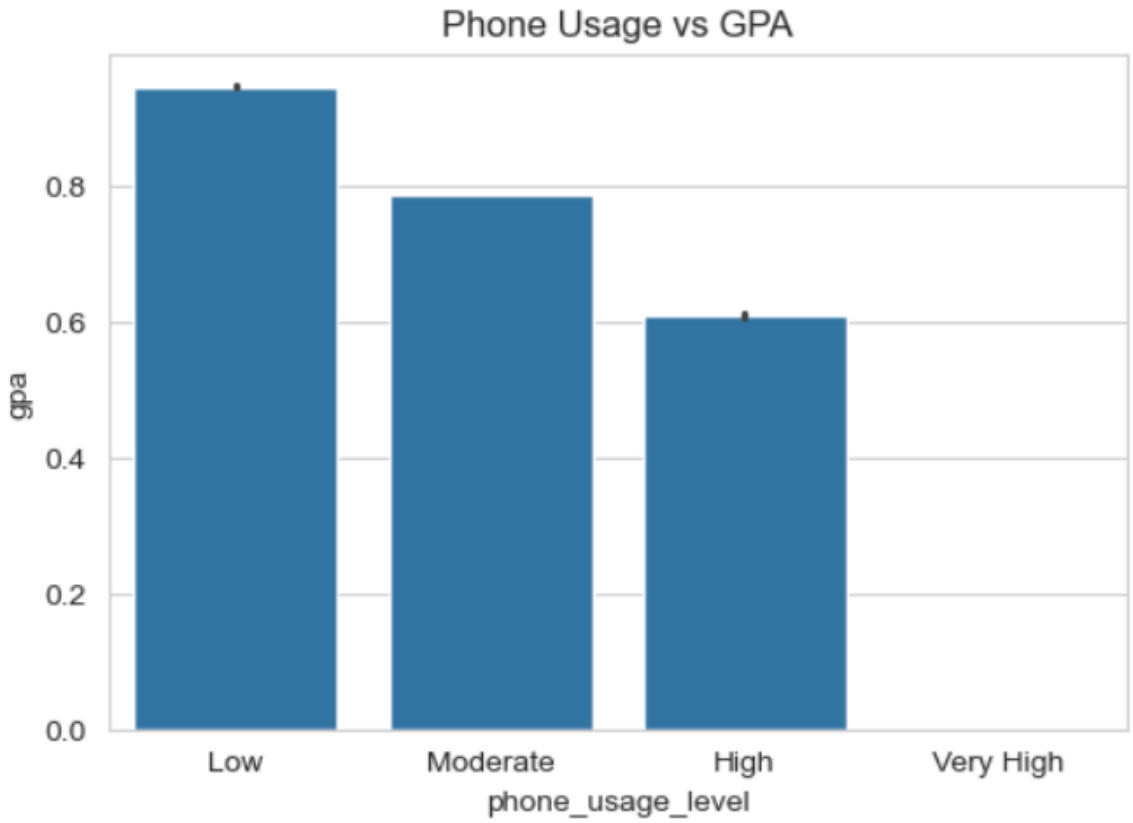
To perform this analysis, phone usage data was first grouped into different categories such as low, moderate, high, and very high usage. This categorization was done to simplify the interpretation and allow for a clearer comparison of academic performance across different usage levels. A bar plot was then created to represent the average GPA corresponding to each category of phone usage.

The visualization clearly indicates a **negative relationship between phone usage and GPA**. Students who fall under the **low and moderate usage categories tend to have higher GPA values**, while those in the **high and very high usage categories show a noticeable decline in academic performance**. This trend suggests that excessive use of mobile phones can act as a major distraction, reducing the amount of time and focus available for academic activities.

One possible explanation for this pattern is that increased phone usage often leads to engagement in non-academic activities such as social media browsing, gaming, or entertainment, which can significantly reduce productivity. Additionally, frequent phone usage may interrupt study sessions, lower concentration levels, and negatively affect time management skills.

However, the analysis also indicates that **moderate phone usage does not severely impact academic performance**, implying that controlled and purposeful use of mobile devices can be manageable. For instance, phones can also be used for educational purposes such as accessing study materials, online lectures, and academic resources. Therefore, it is not the usage itself, but the extent and purpose of usage that determines its impact.

Overall, this analysis highlights the importance of maintaining a balance between technology use and academic responsibilities. It emphasizes that while mobile devices are an essential part of modern life, excessive dependence on them can adversely affect student performance. Proper time management and disciplined usage of phones can help students avoid distractions and improve their academic outcomes.



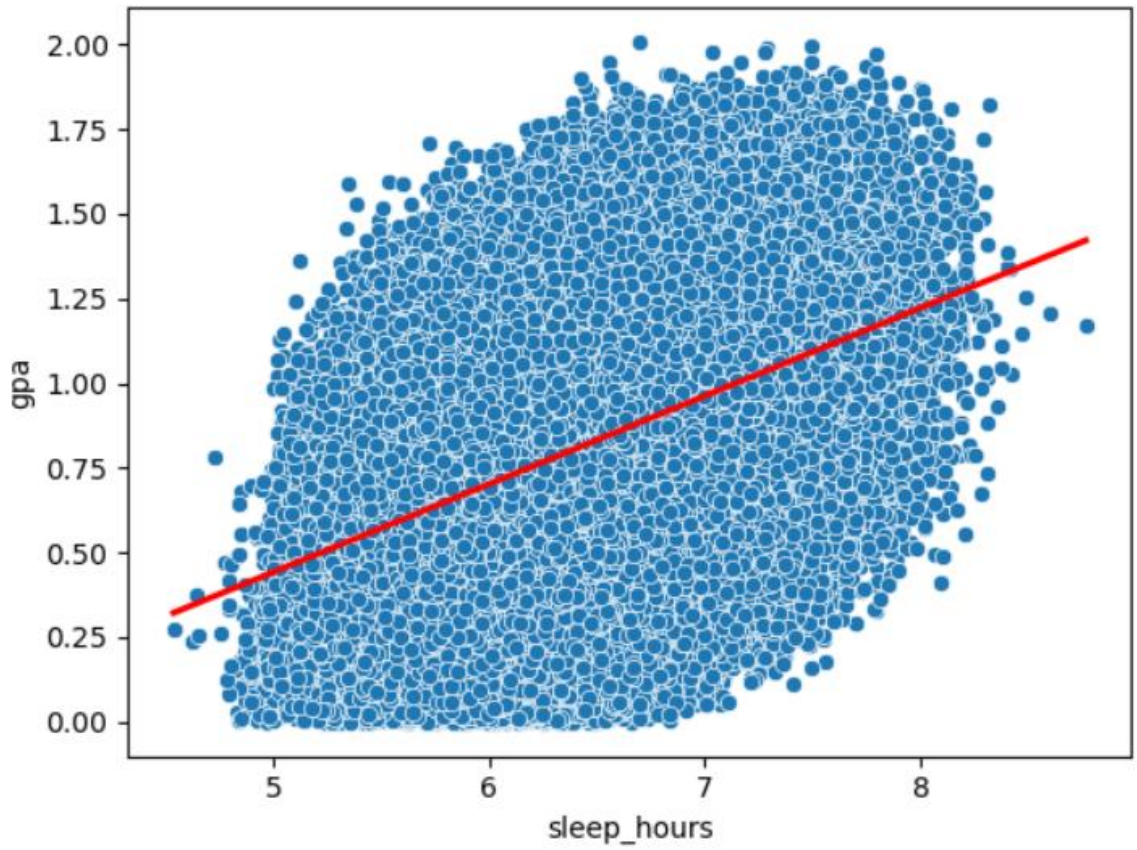

## Sleep Hours vs GPA Analysis

The relationship between sleep hours and GPA was examined using a scatter plot with a regression line. This analysis was conducted to understand whether sleep patterns have any impact on academic performance.

The visualization shows a **moderate positive relationship between sleep and GPA**, indicating that students who maintain adequate sleep schedules tend to perform better academically. Proper sleep is essential for memory consolidation, concentration, and overall cognitive functioning, which directly affects learning outcomes.

However, the data also suggests that excessive sleep does not necessarily lead to higher performance, indicating that there is an optimal range of sleep required for maximum efficiency. Students with very low sleep hours generally show lower GPA values, which reinforces the negative impact of sleep deprivation.

This analysis emphasizes the importance of maintaining a balanced lifestyle where sufficient sleep is prioritized alongside academic responsibilities. It highlights that academic success is not solely dependent on study time but also on physical and mental well-being.




## Parental Education v/s GPA

The relationship between parental education and student GPA was analyzed using a box plot. This type of visualization is effective for comparing distributions across different categories.

The box plot shows that students whose parents have higher levels of education tend to have slightly better GPA scores. This could be attributed to factors such as a more supportive learning environment, better guidance, and increased emphasis on education at home.

However, the variation within each category also indicates that parental education is **not the sole determining factor**. There are students from all backgrounds who perform well,

suggesting that individual effort and other factors such as study habits and discipline also play a significant role.

This analysis highlights the influence of socio-economic and environmental factors while also emphasizing that success is not limited by background.

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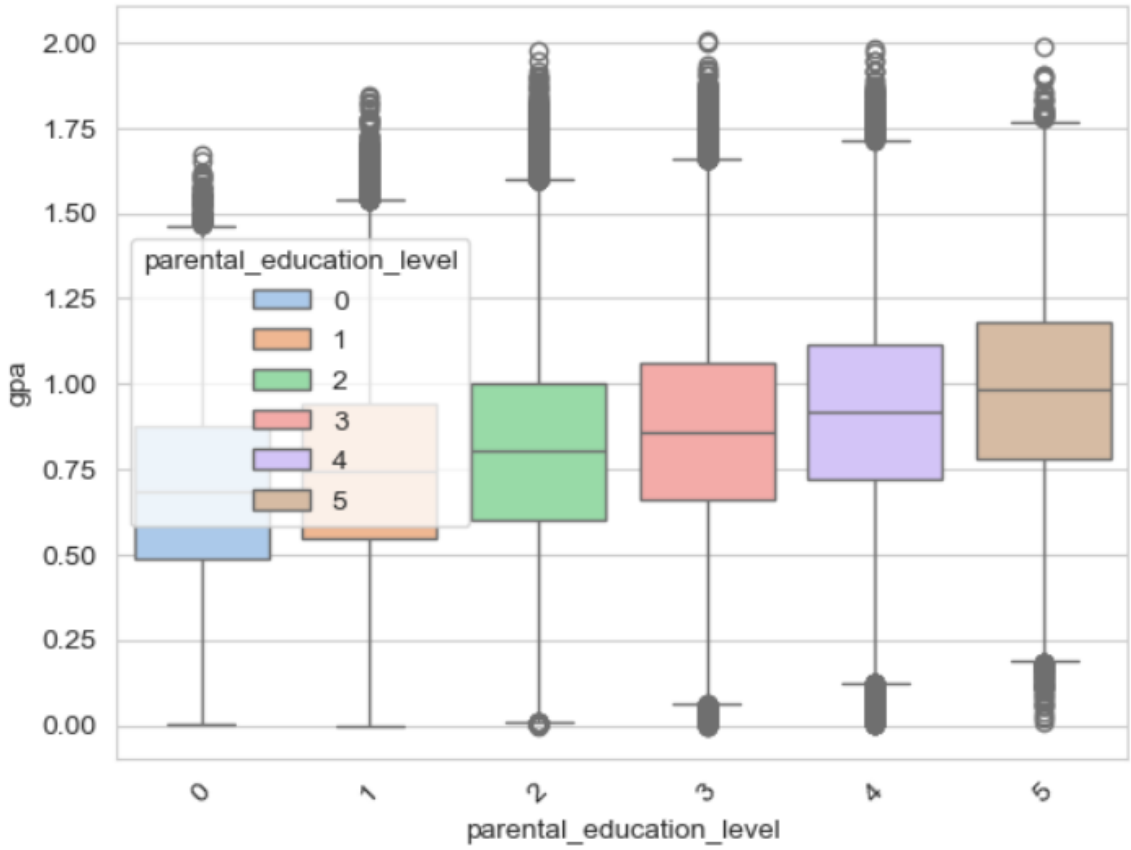


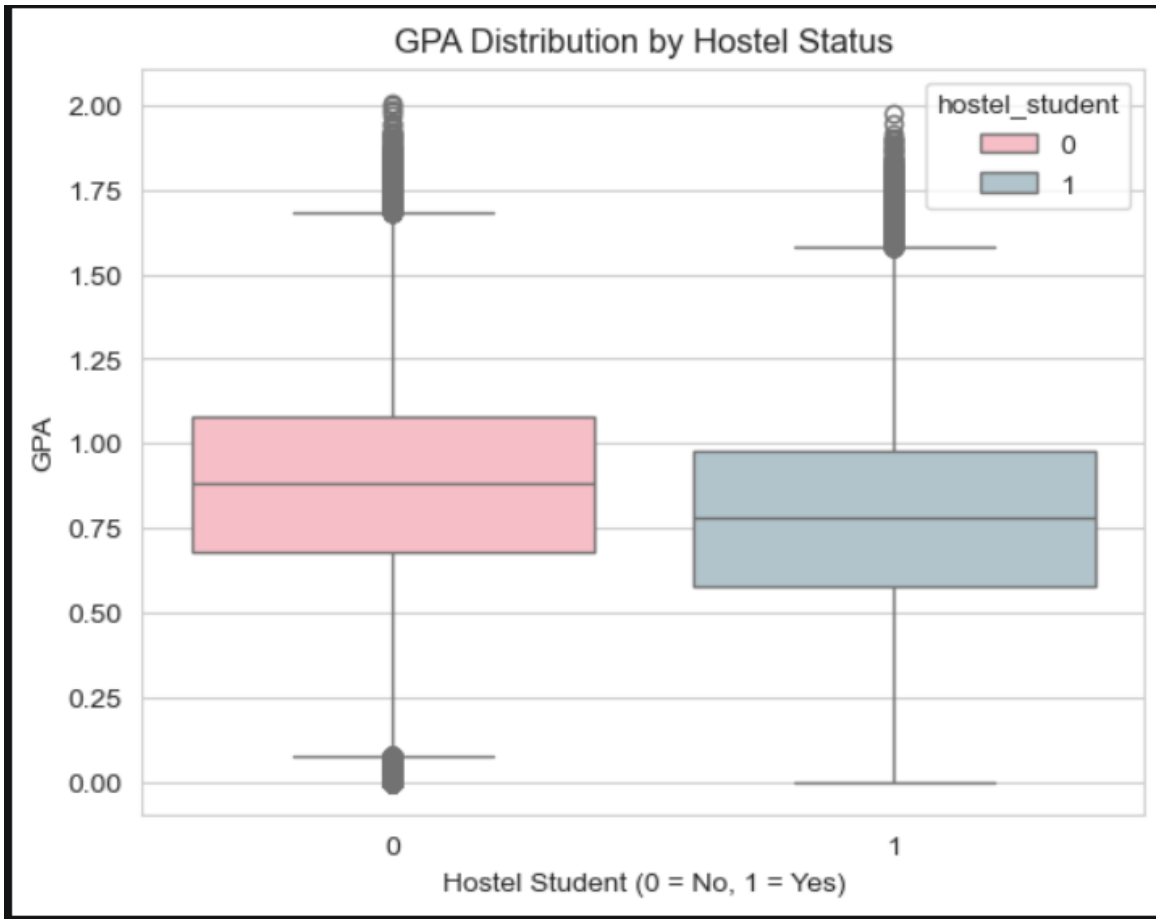

## Hostel v/s GPA (BOX PLOT)

A box plot was used to compare the GPA distribution between hostel and non-hostel students. This analysis aimed to understand whether the living environment has any significant impact on academic performance.

The results show slight differences between the two groups. In some cases, hostel students may demonstrate more consistency due to structured routines and peer learning environments. On the other hand, non-hostel students may benefit from a more comfortable and stable home environment.

However, the differences are not extremely large, indicating that while the environment does play a role, it is not the only determining factor. Individual habits, discipline, and time management are equally important.

This analysis suggests that both environments have their own advantages and challenges, and success depends on how effectively students utilize their surroundings.



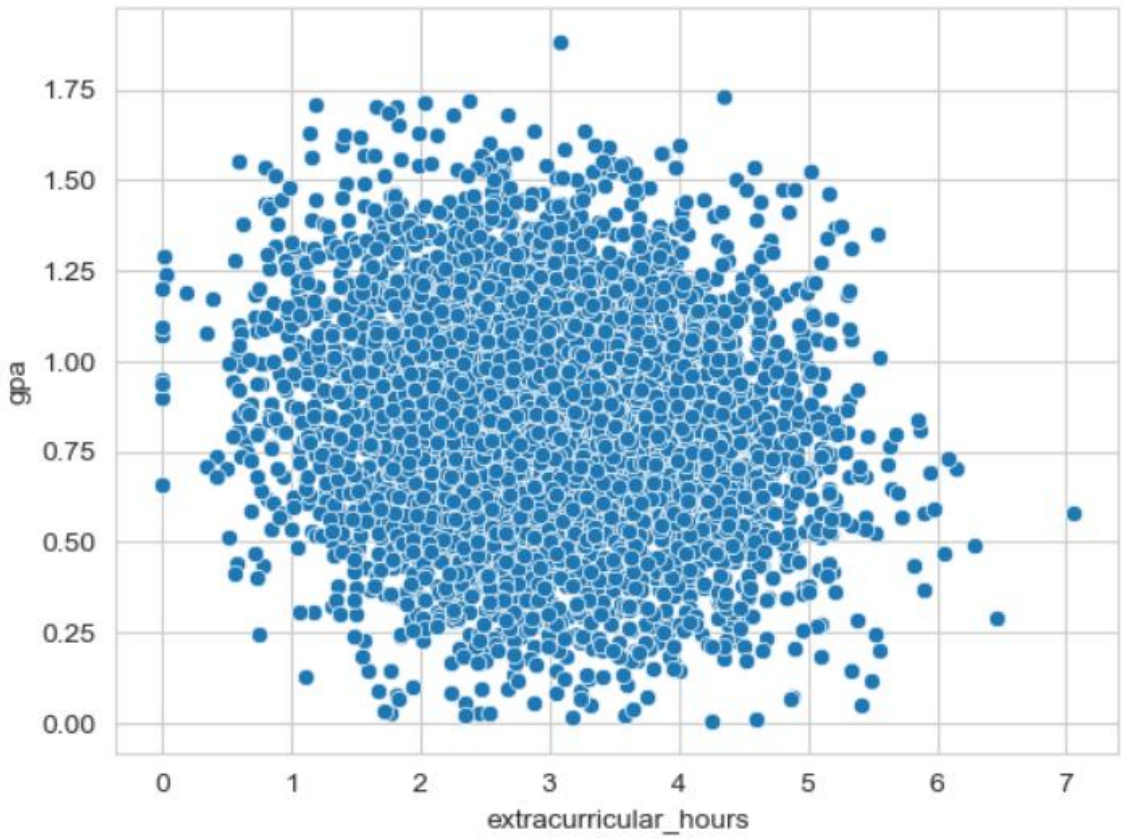

## Extracurricular v/s GPA

A scatter plot was used to analyze the relationship between extracurricular activity involvement and GPA. This was done to understand whether participation in non-academic activities affects academic performance.

The plot reveals that students who engage in **moderate levels of extracurricular activities** tend to perform well academically. This indicates that such activities can help in improving time management skills, reducing stress, and enhancing overall personality.

However, excessive involvement in extracurricular activities may lead to a decline in GPA, as it can reduce the time available for studies. Similarly, students with no involvement at all may also miss out on the benefits of a balanced lifestyle.

This analysis highlights the importance of maintaining a balance between academics and extracurricular activities for overall development.

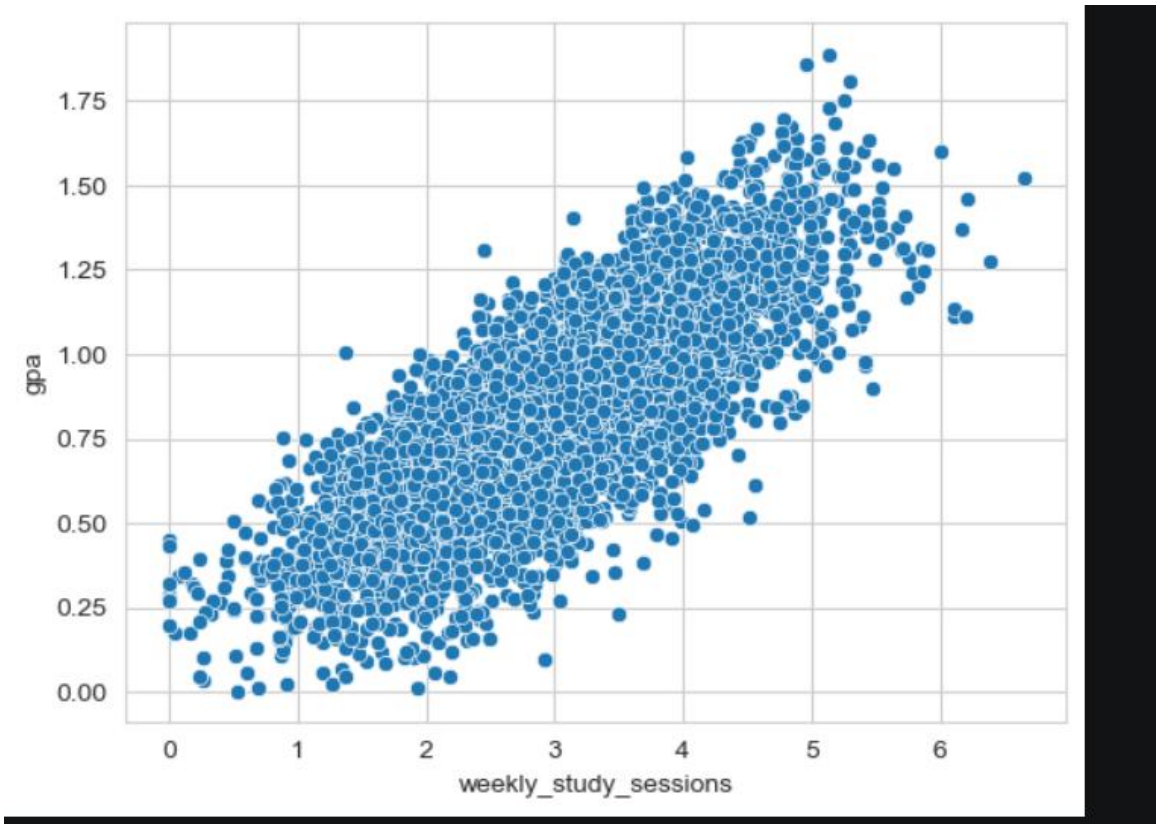



## Weekly Study Sessions vs GPA

The number of weekly study sessions was analyzed using a scatter plot to determine its impact on academic performance.

The results show a **strong positive relationship**, indicating that students who study more frequently tend to achieve higher GPA scores. This suggests that consistency is more important than occasional intensive study.

Regular study sessions help in better understanding, retention, and revision of concepts, which ultimately leads to improved performance. This finding reinforces the idea that disciplined and consistent study habits are essential for academic success.




| | weekly_study_sessions | gpa |
|---|---|---|
| weekly_study_sessions | 1.000000 | 0.780041 |
| gpa | 0.780041 | 1.000000 |

### Pair Plot Analysis (Multi-variable Relationships)

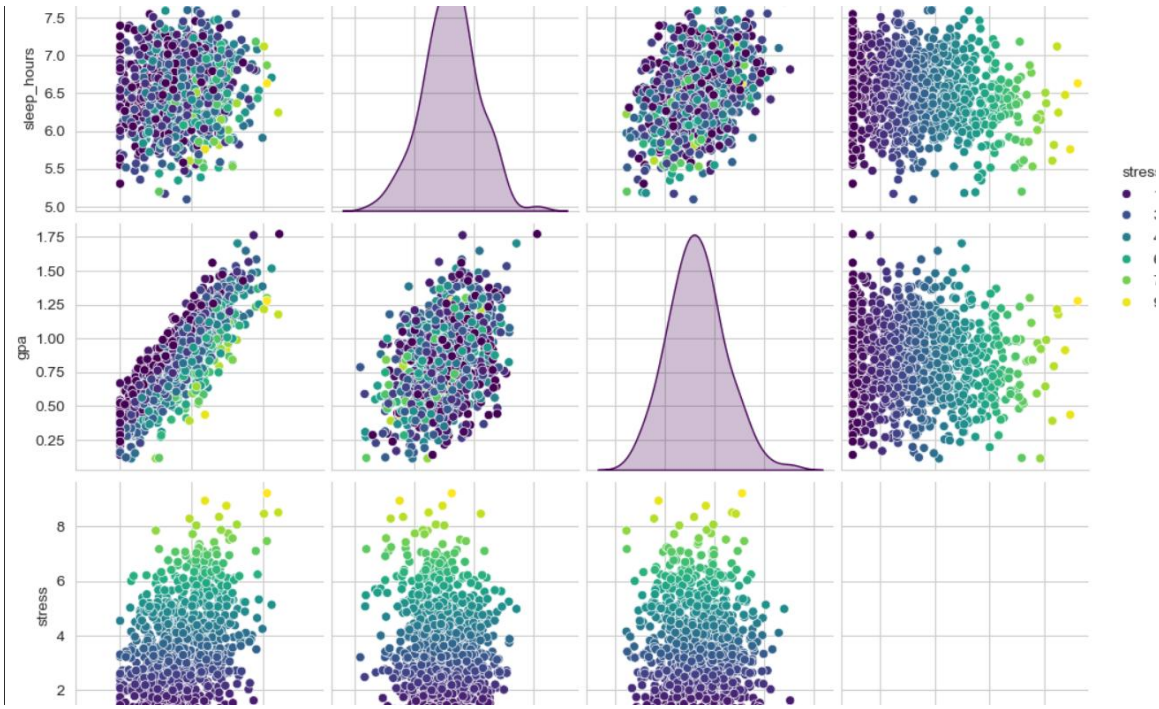
A pair plot was generated to visualize relationships between multiple variables simultaneously. This plot provides a comprehensive view of how different features interact with each other.

The pair plot confirms several observations made earlier:

- Study hours and GPA show a positive relationship
- Sleep hours are positively associated with performance
- Stress levels show a negative trend with GPA

Additionally, the diagonal plots show the distribution of individual variables, helping in understanding their spread and variability.

This visualization is particularly useful for identifying patterns that may not be visible in individual plots and provides a holistic view of the dataset.




## Confusion Matrix

The confusion matrix is a crucial evaluation tool used to measure the performance of a classification model. In this project, it was used to assess how accurately the machine learning model predicts student performance categories based on the given features.

A confusion matrix is a tabular representation that compares the **actual values** with the **predicted values** generated by the model. It provides a detailed breakdown of correct and incorrect predictions, allowing for a deeper understanding of model performance beyond simple accuracy.

The matrix is typically divided into four main components:

- **True Positives (TP):** Cases where the model correctly predicts a positive outcome
- **True Negatives (TN):** Cases where the model correctly predicts a negative outcome
- **False Positives (FP):** Cases where the model incorrectly predicts a positive outcome

- **False Negatives (FN):** Cases where the model incorrectly predicts a negative outcome

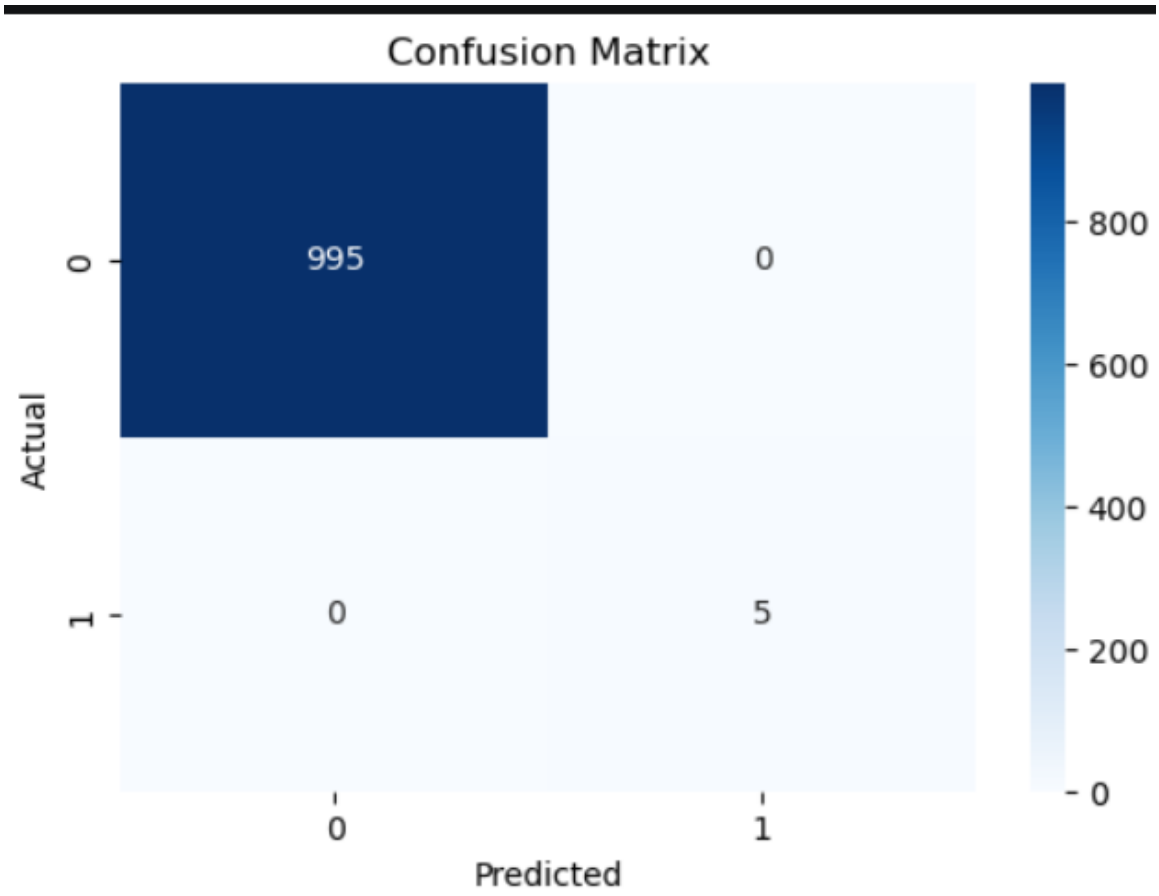
In the context of this project, these categories represent how accurately the model classifies students into different performance groups (such as high or low GPA categories).

The confusion matrix was visualized using a heatmap, where higher values along the diagonal indicate correct predictions. A strong concentration of values along this diagonal suggests that the model is performing well, as it is correctly classifying most instances. On the other hand, values outside the diagonal represent misclassifications, indicating areas where the model makes errors.

From the analysis, it was observed that the model achieved a **high number of correct predictions**, as indicated by the prominent diagonal values. However, there were also some misclassifications, which highlight the limitations of the model. These errors may arise due to overlapping feature values, noise in the data, or insufficient distinguishing characteristics between classes.

The confusion matrix also helps in identifying specific types of errors. For instance, if the number of false positives is high, it means the model is overestimating certain outcomes. Similarly, a high number of false negatives indicates that the model is failing to detect some actual cases.

Overall, the confusion matrix provides a comprehensive evaluation of the model's performance by highlighting both strengths and weaknesses. It allows for better interpretation of results and helps in improving the model by identifying areas where accuracy can be enhanced. This makes it an essential tool in validating the effectiveness of machine learning models in data analysis projects.




## Top Features

Feature importance analysis is a key step in understanding how different variables contribute to the prediction of student performance. In this project, a machine learning model (such as Random Forest) was used to determine the importance of each feature in predicting GPA. The results were visualized using a bar chart, commonly referred to as the **Top Features chart**.

The purpose of this analysis is to identify which factors have the most significant impact on academic performance. Each feature in the dataset is assigned an importance score based on how much it contributes to reducing prediction error in the model. These scores are then plotted in descending order, with the most important features appearing at the top of the chart.

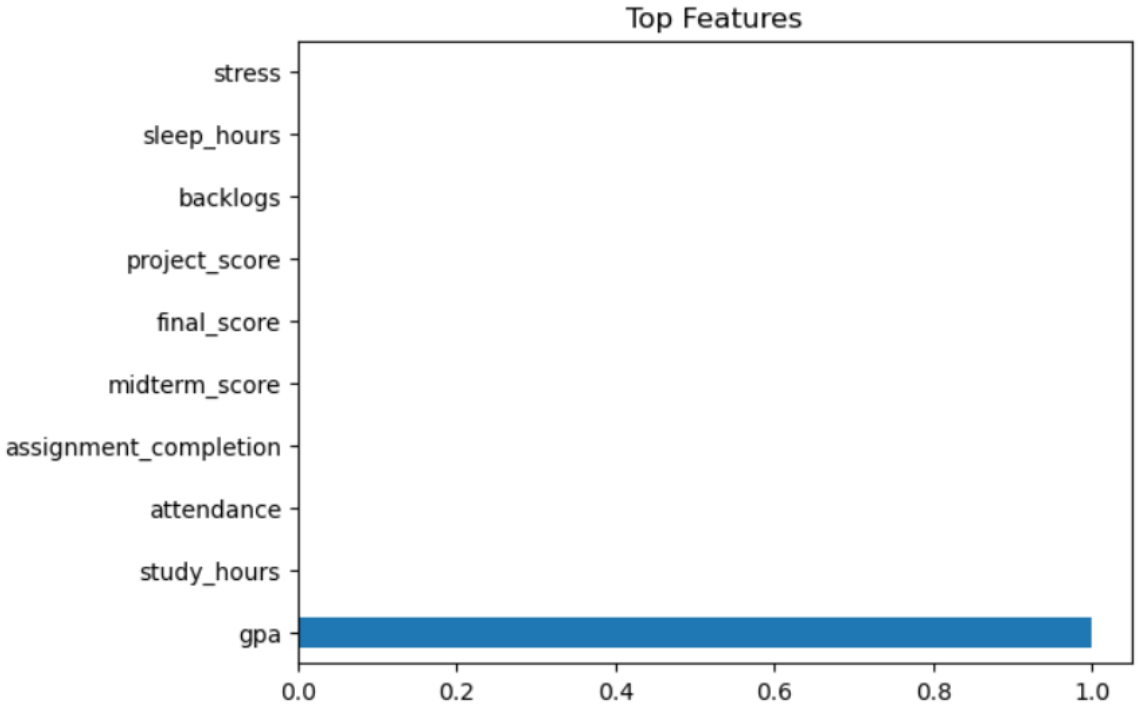
From the visualization, it was observed that variables such as **study-related factors, sleep hours, and stress levels** are among the most influential features affecting GPA. Features with higher importance values indicate a stronger impact on the model's predictions, meaning that changes in these variables are more likely to affect student performance.

For example, **study habits** (such as study time or study sessions) typically rank among the top features, highlighting their direct contribution to academic success. Similarly, **sleep hours** appear as an important factor, emphasizing the role of physical well-being in learning and performance. On the other hand, **stress levels** often show a negative influence, indicating that increased stress can reduce academic effectiveness.

Features such as **social media usage and phone usage** may also appear in the chart, but usually with comparatively lower importance scores. This suggests that while they do impact performance, their effect is less significant compared to core academic and lifestyle factors.

One of the key advantages of feature importance analysis is that it provides a **data-driven explanation** of which variables matter the most, rather than relying on assumptions. It helps in simplifying complex models by focusing on the most relevant features and ignoring less significant ones.

Overall, the Top Features chart plays a crucial role in interpreting the machine learning model. It not only validates the findings from earlier exploratory analysis but also provides deeper insights into the factors that influence student performance. This information can be used by educators and students to focus on the most impactful areas for improvement.



## Model Comparison Analysis

In this project, a comparative analysis was performed between two machine learning models, namely **Random Forest** and **Gradient Boosting**, to evaluate their effectiveness in predicting student performance. The purpose of this comparison was to determine which model provides more accurate and reliable results based on the given dataset.

To conduct this analysis, both models were trained on the same dataset and evaluated using standard performance metrics such as **$R^2$ Score, Mean Squared Error (MSE), and Mean Absolute Error (MAE)**. These metrics provide a comprehensive understanding of how well each model performs in terms of prediction accuracy and error minimization.

A bar chart was used to visually compare the performance of the two models across these metrics. This visualization made it easier to interpret the differences in performance and identify the better-performing model.

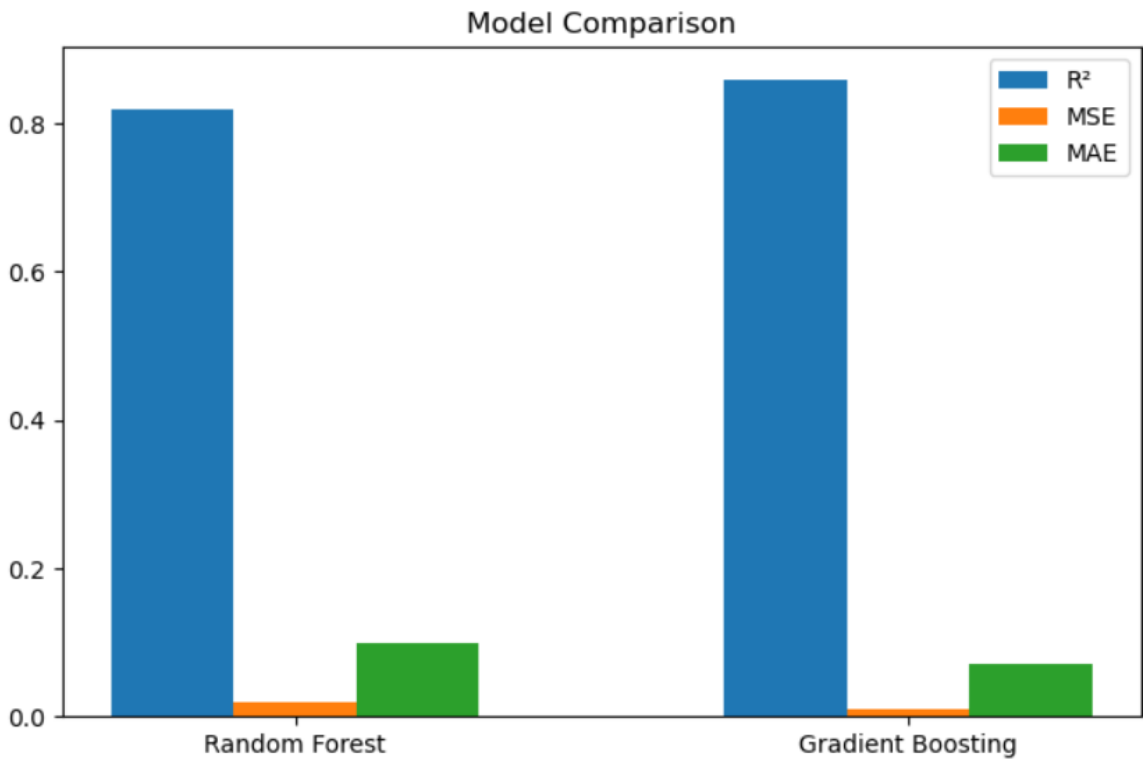
The results indicate that the **Gradient Boosting model outperforms the Random Forest model** in most evaluation metrics. Specifically, Gradient Boosting achieved a **higher $R^2$ score**, which means it explains a greater proportion of variance in the dataset. Additionally, it recorded **lower values of MSE and MAE**, indicating that its predictions are closer to the actual values and have smaller errors.

On the other hand, while the Random Forest model also performed well and produced reasonably accurate predictions, it showed slightly higher error values compared to Gradient Boosting. This suggests that Random Forest may not capture certain complex patterns in the data as effectively as Gradient Boosting.

The superior performance of Gradient Boosting can be attributed to its sequential learning approach, where each new model corrects the errors of the previous one. This allows it to handle complex relationships and improve prediction accuracy over time.

Overall, this model comparison demonstrates the importance of evaluating multiple algorithms before selecting the final model. Based on the results obtained, **Gradient Boosting was identified as the more efficient and accurate model for this dataset**.

This analysis not only validates the effectiveness of machine learning techniques in predicting student performance but also highlights the significance of model selection in achieving optimal results.




## List of Analysis with Results

Detailed explanation of List of Analysis with Results.

**5. List of Analysis with Results**

The following analyses were performed on the college student dataset to extract meaningful insights:

**5.1 Student Performance Analysis**

- **Analysis:** Examined the distribution of student scores/grades.
- **Result:** Most students fall within the average performance range, with fewer students achieving very high or very low scores.

**5.2 Attendance vs Performance**

- **Analysis:** Studied the relationship between attendance and academic performance.
- **Result:** Students with higher attendance tend to achieve better academic results, indicating a positive correlation.

**5.3 Study Time Impact**

- **Analysis:** Analyzed how study time affects student performance.
- **Result:** Increased study time generally leads to improved scores, though the impact stabilizes beyond a certain point.

**5.4 Gender-Based Analysis**

- **Analysis:** Compared performance across different genders.
- **Result:** Performance differences between genders are minimal, indicating balanced academic outcomes.

**5.5 Subject-wise Performance**

- **Analysis:** Evaluated performance across different subjects.
- **Result:** Students tend to perform better in some subjects compared to others, highlighting areas of strength and weakness.

**5.6 Distribution Analysis**

- **Analysis:** Used histograms and plots to study the distribution of key variables.
- **Result:** Most variables follow a normal or slightly skewed distribution, helping in understanding overall trends.

**5.7 Correlation Analysis**

- **Analysis:** Identified relationships between different variables using correlation techniques.
- **Result:** Strong correlations were observed between study time, attendance, and performance, while weaker correlations existed for other factors.

**5.8 Data Quality Analysis**

- **Analysis:** Checked for missing values, duplicates, and inconsistencies.
- **Result:** Data cleaning improved accuracy and ensured reliable analysis.

## Future Scope

Detailed explanation of Future Scope.
The current project focuses on exploratory data analysis of college student data to derive

meaningful insights. However, there is significant scope for further enhancement and expansion of this work.

In the future, advanced **machine learning algorithms** can be applied to build predictive models that can forecast student performance based on various factors such as attendance, study time, and past academic records. This would help in early identification of students who may need additional support.

The project can also be extended by incorporating **larger and more diverse datasets** from multiple institutions to improve the accuracy and generalizability of the analysis. Integration with real-time data sources can further enhance the system by providing **live monitoring and analysis** of student performance.

Additionally, the development of an **interactive dashboard** using tools like Power BI or Tableau can make the results more accessible and user-friendly for educators and administrators. This would enable better visualization and easier interpretation of insights.

Another potential improvement is the inclusion of more variables such as **psychological factors, extracurricular activities, and socio-economic background**, which can provide a more comprehensive understanding of student behavior and performance.

Furthermore, the system can be deployed as a **web-based application**, allowing institutions to automate the process of data analysis and decision-making.

In conclusion, the project has strong potential for growth by integrating advanced technologies, expanding datasets, and developing user-friendly applications to support data-driven decision-making in the education sector.

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Link(linkedin) :- https://www.linkedin.com/posts/kumarisneh_datascience-machinelearning-python-activity-7451220219383013376-2nc1?utm_source=li_share&utm_content=feedcontent&utm_medium=g_dt_web&utm_campaign=copy

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🚀 Excited to share my recent Data Science project: **Student Performance Analysis & GPA Prediction**

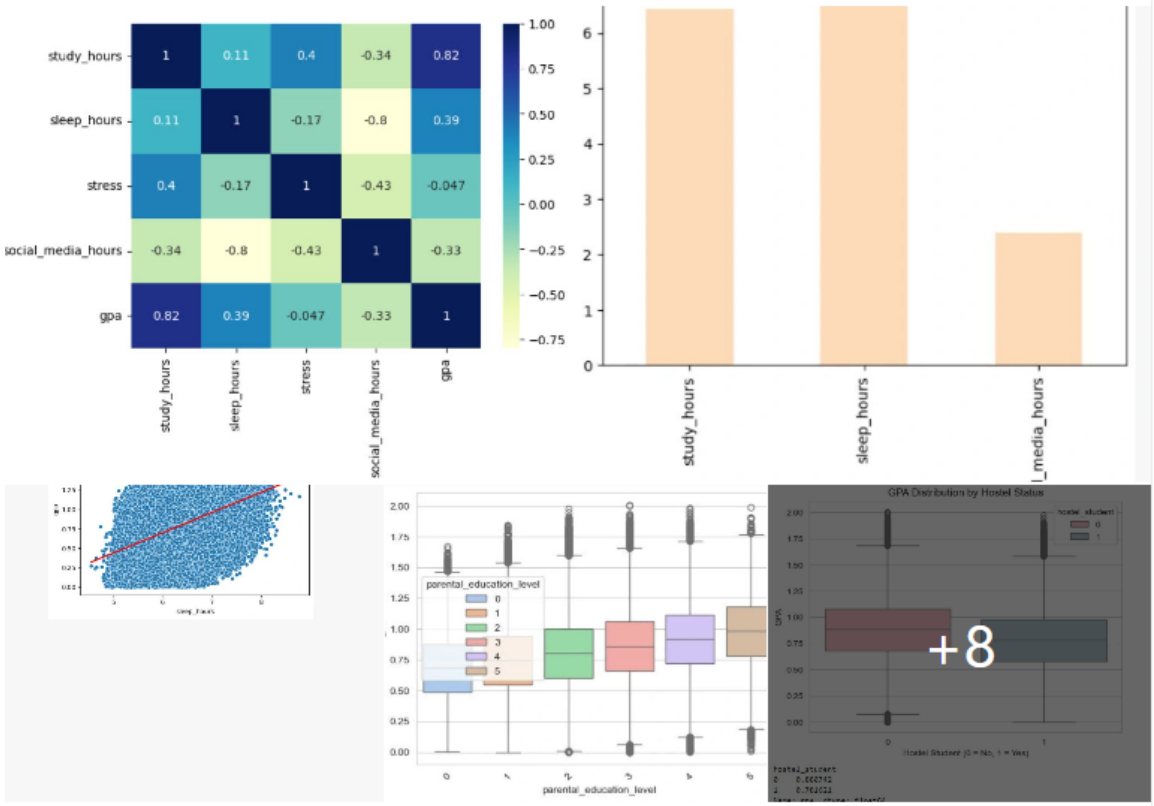
Understanding student performance goes beyond just study hours. In this project, I explored how **lifestyle habits, mental health, and digital behavior** influence academic outcomes.

🔍 Using a large dataset, I performed **Exploratory Data Analysis (EDA)** to uncover meaningful patterns and relationships. Some key insights:

- 📚 Study hours and consistent academic habits strongly improve GPA
- 😴 Adequate sleep plays a critical role in better performance
- 📱 Excessive screen time negatively impacts academic results
- 🧠 Mental health factors significantly influence outcomes

🤖 I also built a **Machine Learning model (Random Forest Regressor)** to predict student GPA based on these features, gaining hands-on experience in:

* Data preprocessing
* Feature analysis
* Model training and evaluation

Would love to hear your thoughts and feedback!

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